

MOST HAPPENING, FACT-FILLED CURRENT AFFAIRS FROM AROUND THE WORLD

Hand gesture recognition for mobile phones now

Mobile phones could soon be turned into hand gesture recognition (HGR) devices following a technological breakthrough. Software developed by researchers at the University of Warwick would allow the 2D cameras in everyday consumer electronic products, such as mobile phones and tablet computers, to become HGR devices in uncontrolled environments with both natural and artificial light. A patent application has been filed to Patent Cooperation Treat (international) Stage.

Since the early 1990s, HGR has been intensively researched in both academic and industrial communities, but until now there have been no successful academic theories or commercial products that have performed robust HGR in uncontrolled environments.

Existing HGR technologies require infrared or 3D cameras and will only work in artificial light, and so have been limited to the gaming industry and smart TVs. The new technology, which can also recognise both gesture and posture, is more economical, practical to use and opens up the possibilities of HGR being used in the automotive, health and consumer electronics sectors.

PV cells to recycle thermal energy

Researchers at the University of Buffalo in their latest research paper titled 'Broadband absorption engineering of hyperbolic metafilms patterns' have proposed the use of multi-layered waveguide taper array that opens the door for newer and better possibilities in the field of thermal energy, solar power and energy conversions. The research is spearheaded by lead researcher Qiaoqiang Gan and predicts



Hand gesture recognition for video games

better times to come in the aforementioned fields.

The waveguide tapers are micro-chip components that allow chips to trap and absorb light of each frequency, thereby catching a rainbow of wavelengths or broadband light. The researchers had predicted earlier that multi-layered tapers could absorb light more efficiently, only now they have successful experimental results to prove the same.

During the research, multi-layered taper made up of ultra-thin layer of metal, semiconductors and/or insulators was employed. The taper, named hyperbolic metamaterial, has the ability to absorb light in between pairs of layers and adjusting the geometry and thickness of the taper results in absorbing various frequencies.

The research opens up new avenues in the field of thin-film solar tech-



The waveguide tapers absorb light of each frequency to catch a rainbow of wavelengths or broadband light